

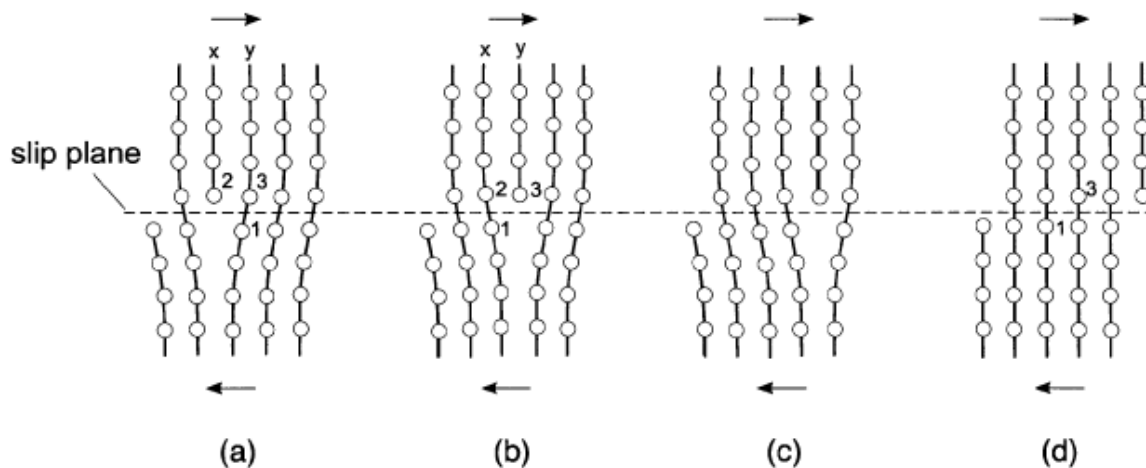
APS110 – Lecture 17

See hand written notes for drawings.

Imperfections

Recall:

- 0D → point imperfections
- 1D → linear imperfections (dislocations)
- 2D → grain boundaries
 - Grain size reduction as a strengthening mechanism
 - Different types (meet at close angle vs. high angle, etc.)
 - Free surfaces → another 2D imperfection
 - Something to be aware of, but we don't really use them for strengthening



3-Dimensional Imperfections ("Volume imperfections")

- **Larger than the atomic scale**
- Ex. air bubble in glass, a large pore in a metal casting
- Recall that a point imperfection is a single atomic change in a lattice (ex. one single atom missing)
 - A volume imperfection is on a much larger scale than this → a whole cluster of atoms, maybe even a thousand, a million atoms missing, to form a pore
- 3D imperfections examples: pores, 2nd phase particle
 - We will revisit second phase particles later, when we talk about phase equilibria
- Ex. Fe₃C in steel (2nd phase particle)
 - In steel you have BCC iron, with a little bit of carbon dissolved (0.022 wt% C)
 - This is only a tiny bit of carbon, where is the rest of it?
 - The rest of the carbon comes out of solution to form Fe₃C along the grain boundaries
 - There can be other structures (plates, layers) that can form within the grains

- This Fe_3C has different crystal structure, different properties from the parent lattice
 - This is a 3D imperfection
 - Similar to grain boundaries, 2nd phase boundaries impede dislocation movement
 - Ex. Al – AA6061 T6
 - o 6061 → describes the specific chemistry of Aluminum
 - o T6 → specific heat treatment (precipitation heat treatment)
 - Very controlled way of introducing second phase boundaries
 - These are the a major mechanism of strengthening for this aluminum alloy
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POLYMERS

- Stress-strain curve looks different from metals
 - o There is an increase, then a peak, but then the curve drops off and the polymer does not fracture right away
 - o How does continued load bearing occur?
 - o Just like with metals, necking occurs, which causes the decrease in stress after the peak
 - But then, the polymer can continue bearing load without fracture
 - The curve even starts to increase again → how is it getting stronger?
- Not crystalline like metals → held together by covalent bonds
 - o Most of the strength is actually coming from weaker bonds between the molecules
- Different factors with polymers (as compared to metals, ceramics)
 - o Time-dependent deformation
 - o Temperature-dependent deformation
- Silly putty example
 - o Viscous flow of putty → deforms faster when warm
 - o Loaded at a high strain rate, its elastic (bouncing)
 - Behaves completely different from loading at low strain rate
 - o Can be brittle, when loaded at a high strain rate (tears, no longer extremely ductile)
 - Classic brittle fracture surface, no plastic deformation
- The key to understand the behaviour of polymers is to look into the **microstructure**
- Water bottle
 - o At the molecular level, there is a jumble of very long molecules
 - o These molecules would be made up of long strings of carbon, covalently bonded to each other, then bonded to hydrogen atoms → hydrocarbon
- We can make lots of changes to this microstructure to influences the produce different properties
- Polymers
 - o Often hydrocarbons (not always)

o Building blocks are '**mer units**'

- What is the smallest repeating unit that we can use to describe this molecule? → see hand written notes
- Why do we use the two-carbon unit as opposed to a single carbon unit?
 - This is because we start with **ethene**
 - When we synthesize this polymer, we start with ethene then we break the double bond which makes the molecule a free radical
 - Thus, this mer unit will go off and react with another unit and they will start adding onto each other
- We call the large molecule **POLYETHYLENE**, or **POLYETHENE**

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